Venkata Lakshmi Parimala Pasupuleti

Senior Full Stack Engineer · AI-Assisted Development (Claude Code) · Real Estate SaaS · LLM API Integration · End-to-End Ownership · 9yr Production

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PROFESSIONAL SUMMARY

Senior full stack engineer with 9+ years in real estate SaaS -- I use Claude Code as a core daily development tool and have built production AI workflows on top of it. At Entrata, I owned end-to-end delivery of business-critical systems for 100K+ housing units across 120+ Agile sprints -- real estate data with compliance consequences, zero tolerance for correctness errors. My open-source AI projects apply the same discipline: LeaseI is an 8-agent LangGraph AI platform for real estate workflows (LLM orchestration, domain document retrieval, HITL escalation, CI quality gates); AgentProbe is an evaluation toolkit built to hold AI-generated code to the same standard as hand-written code. Backend: Python/FastAPI/Flask, 9yr production (meets Flask/Node "or similar"). Frontend: Angular 6yr at Entrata, React on projects. I'm drawn to Dealpath because it applies AI to real estate investment decisions -- a domain I know deeply from the operations side, and want to understand from the capital markets side. Portfolio: venkatalakshmiparimala.github.io/Portfolio/about.html

TECHNICAL SKILLS

Languages & Frameworks: Python (9yr production), SQL/PL-SQL, TypeScript, JavaScript, R, Java, Bash/shell scripting, HTML/CSS; FastAPI, Flask, Django (familiarity), Angular, React.js, Spring Boot (Ruby: ramping)

AI & Development Tools: Claude Code (daily), Cursor, AI-assisted development, LLM API integration (Anthropic, OpenAI, Groq), LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, RAG Pipelines, ChromaDB, Prompt Engineering, LLM Evaluation, LLM-as-Judge, Hallucination Detection, HITL, Guardrails, Model Evaluation, OCR, Document Intelligence, Embeddings, NLP

Backend, Data & Infrastructure: REST APIs, Microservices, PostgreSQL (200+ queries optimized, stored procedures, multi-tenant schemas), MySQL, Oracle, NoSQL, SQLite, ETL/ELT Pipelines, Data Modeling, AWS (S3, EC2, RDS, Glue), Azure, GCP, Docker, Kubernetes, RabbitMQ async processing, CI/CD (GitHub Actions), Automated Testing, Regression Testing, Performance Optimization, Observability, Structured Logging, Git, Agile/Scrum, Code Reviews, Technical Mentoring

EXPERIENCE

Entrata -- Multifamily Real Estate SaaS (PropTech)

Sep 2018 -- Aug 2024

Senior Software Engineer -- Full Stack, Real Estate Data Pipelines & Compliance · Lehi, UT (Remote)

- **End-to-end ownership for 8 years in high-growth real estate SaaS** -- sole owner of Entrata's HUD/LIHTC/HCV income-limit and eligibility system for nearly 7 years: annual ingestion of federal income-limit data (few thousand source rows expanded to ~180,000 domain records), eligibility and rent calculations, and recertification workflows for 100K+ housing units. Real estate data with compliance consequences -- every release required pre-import baseline validation, reconciliation, and rollback preparation. Zero compliance violations across full ownership period.
- **Full stack feature delivery and code quality** -- shipped 20+ full-stack production modules (Python, Angular/JavaScript, PostgreSQL) across 120+ Agile sprints; 300+ peer code reviews with focus on correctness, edge cases, and long-term maintainability; Docker microservices redesign (50% scalability improvement, 30% deployment time reduction); eliminated 80% of manual operational workload through Python/Bash automation pipelines.
- **Backend and data engineering at scale** -- RabbitMQ async processing with retry logic and idempotency guarantees for gross-rent updates across thousands of tenant records; optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, and partitioning; owned complex schemas for eligibility, financial, and document workflows across multi-tenant production environments.
- **Cloud, integrations, and production reliability** -- led AWS S3 document migration with Libsodium-encrypted ETL pipelines (100% PHI compliance, zero security incidents); integrated enterprise workflows with REST APIs, third-party data providers, and multi-tenant RDBMS systems; mentorship of 5+ engineers (code reviews, pairing, career guidance); built internal debugging tools reducing debugging time by 40%.

George Mason University

Aug 2024 -- May 2026

Research Associate & Graduate Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) on WHO and UK Road Safety datasets across 190+ countries (R2=0.977 regression accuracy); applied Isolation Forest to automate data integrity checks across 3 research pipelines; designed Tableau, ggplot2, and Quarto dashboards improving research comprehension 40% for faculty leadership; mentored 600+ students in Python, SQL, and data engineering.

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer -- Agentic Systems & LLM API Integration

- Built an agentic mission-planning system using Claude Code as the primary development tool: LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing), RAG layer over FAA Part 107 regulatory corpus with cited sources, automated validation guardrails, and CI-integrated testing gate -- sole technical owner from specification through production delivery.

Entrata India -- Pune

Jan 2017 -- Sep 2018

Associate Software Engineer -- Python · PostgreSQL · Full Stack

- Built property management and compliance modules (Python, PostgreSQL, Angular); income-limit import pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints; 56 state-specific housing certification document templates.

PROJECTS

Leasel: AI-Powered Real Estate Workflow Platform -- LLM API Integration + Production Quality Gates

LangGraph · LLM APIs · Real Estate AI · Claude Code · FastAPI · CI/CD

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · SQLite · OCR · pdfplumber · GitHub Actions | github.com/venkatalakshmiparimala/Lease_complaince_rag

AI-powered platform for real estate compliance workflows -- the same architectural patterns Dealpath applies to investment decision workflows. **LLM API integration in production:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → HITL Escalation) with multi-surface shared evidence model; multi-provider LLM routing across Anthropic, OpenAI, and Groq; 93.6% precision and 87.9% F1 across 1,240 real estate compliance test cases. **Production quality gates:** CI-integrated regression gate blocking deploys on score degradation; multi-trial sampling harness; LLM-as-judge evaluation; hand-rolled TF-IDF RAG (no framework dependency -- deliberate for reliability). Built with Claude Code as the primary development tool throughout. Core design rule: LLM reasons; human or deterministic logic makes every final regulated call.

AgentProbe: Open-Source AI Evaluation Toolkit -- AI-Generated Code Quality Gates

Claude Code · CI/CD · LLM Evaluation · FastAPI · Open Source

Python · FastAPI · React.js · TypeScript · ChromaDB · GitHub Actions · Anthropic / OpenAI / Groq APIs | github.com/venkatalakshmiparimala/AgentProbe

Production AI evaluation and quality framework -- built with Claude Code to hold AI-generated code to the same standard as hand-written code before it ships. **Quality gate:** 5-metric automated evaluation engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI pipeline (GitHub Actions) blocking quality degradation, multi-trial regression detection. **Platform:** 15-endpoint FastAPI backend, React.js + TypeScript frontend, multi-provider LLM routing, structured observability. Demonstrates the same code-review discipline for AI-generated output that Dealpath's JD describes: "evaluate AI-generated code by the same standard as hand-written code."

AI on FHIR: Flask + React Full Stack NLP Platform

Flask · React · LLMs · Full Stack · Docker

Python · Flask · React · Docker · spaCy · BERT · Embeddings · FHIR API | github.com/venkatalakshmiparimala/AI_on_FHIR

Full stack Flask + React platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings -- reduced manual clinical retrieval time ~60%; enforced HIPAA-compliant RBAC across structured medical records. Demonstrates the Flask backend + React frontend stack Dealpath uses.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2013--2017